

Agenda Today

- Part List and morphologization
- Coming up with tasks and dividing them among us
- Discussing problem formulation and initial Methodology

Morphology

Luis

- Microcontroller: ESP32 (no specific variants), STM32, Arduino, Raspberry Pi Pico)
- Motors: IDC 187rpm
- Drivers: 3.6A
- Blower: BLDC Luis 47db,
- Heat recognition: Temp Sensor, Thermistor
- Motion Blockage: IR Sensor, Limit Switches, Optical Sensor for Bottom
- Circuit Safety: Programmable Fuse, Varistors,

Gabriele

- Vacuum System and collector: Centrifugal Blower 10 types,
- Collector: Need Separator Possibly, cyclone Separator, gravity settling chamber, bag house separation,
- Cleaning mechanism: need brushes possibly -> reuse driving motor
- Battery Capacity: Voltage Divider ADC Circuit, OPAMP
- Collector Capacity indicator: IR Sensor, Software Solution time schedule

Jessica

- Driving Motor: Direct Drive, timing belt
- Steering mech: Tricycle, Omnidirectional, Servo
- Nr of wheels: 3 or 4 / Tracks
- Types of Motors for motion: NDC, STEPPER, BLDC
- Types of wheels -> threaded wheels / Silicon/ mehanum wheels / omni / Fome field
- Wheel to shaft coupler: Set screw adapter

Alain

- SLAM: LiDar -> min raspberry pi 4
- Bump and turn: 1 to 4 Limit Switches

- Possible infrared sensors / ToF sensors
- Northstar method
- Speakers possibly → more pins more suck

Javier

- Velocity sensor: magnetic sensor / mechanical sensor
- Accelerometer and gyroscope: IMU thing or whatever
- Navigation: Ultrasonic /
- Current Sensor possibly to know if its stuck.
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Driving Motors

Name	Price	Voltage	Current	Torque	RPM	
Luis Motor	241kr	12V		196mN/m	87	

Motor Drivers

Name	Price	Amnt				
H bridge						

Blowers

Name	Price	Noise	Voltage	Pressure	Airflow	

Parts we need:

Motor for Driving and Steering: Jessica

Motor for Vacuuming: Gabriele

Motor for Brushes: Gabriele

Microcontroller: Lukas

Motordrivers: Gabriele Jessica

Heat recognition: Luis, Alain

Circuit Safety things: Luis, Alain

Battery Capacitor: Gabriele

Collector Indicator: Gabriele

Wheels: Jessica

Wheel to shaft coupler: Jessica

Shaft to motor coupler: Jessica

Navigation Sensors: TOF, Servo, IR Sensor Alain Luis

Sensor for being Stuck:

Gyroscope: Alain, Javier

Accelerometer: Alain, Javier

NOTES FOR ME:

- Make plan on github project so theres deadlines
- Proofread problem formulation
- Make partlist in octopart or sum
- Write more on user and market analysis
- Research microcontroller we want

Lets decide

- Navigation: Full lidar or slam is out the window. -> ToF Sensor / Servo / IR Sensor
- 3 wheeled two motors for each of the act wheels
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Tasks to do

- Step Down Converter
- App development
- Navigation Algo Simulation
- Fake Lidar Buildup